SPENDWISE – SMART EXPENSE MANAGER
A Command Line Based Expense Tracking System

Project Report

Submitted as part of the requirements for

VITyarthi's Evaluated Course Project (BYOP)

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1. Introduction

In today's time, managing personal finances has become very important, especially for students and young individuals. Most people spend money daily but do not keep a proper track of where it is going. Because of this, they often end up overspending or failing to save.

SpendWise is a simple command-line based expense manager that helps users record and analyze their daily expenses. The main idea behind this project was to create something practical and actually usable instead of just making a basic project for submission.

This project focuses on helping users become more aware of their spending habits and take better financial decisions over time.

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2. Objective

The main objectives of this project are:

- To create a simple tool for tracking daily expenses
- To help users manage their monthly budget
- To provide basic insights into spending habits
- To improve financial awareness among users
- To implement real programming concepts in a practical way

3. Problem Statement

Many people do not track their expenses properly, especially students. They spend money on small things regularly but do not realize how much it adds up over time.

There are existing applications for expense tracking, but many of them are either too complex or not convenient for quick usage. Also, most users do not need very advanced features, they just need something simple and effective.

So, the problem is to design a lightweight and easy-to-use system that allows users to track expenses and understand their spending patterns without much complexity.

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4. Methodology / Approach

The project is developed using Python and follows a modular approach. Different functionalities are divided into separate files to keep the code organized.

The application runs in a command-line interface where users interact through menu options. The main components of the project include:

- Authentication system for login and signup
- Expense manager for handling all operations
- Data handler for reading and writing files

User data and expenses are stored in JSON format. Each user has their own separate data file, which helps in managing multiple users.

For analysis, logic-based methods are used instead of machine learning. The program calculates patterns and gives suggestions based on available data.

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5. Features of the Project

SpendWise includes several useful features:

- User signup and login system with password hashing
- Add, update, and delete expense records
- View all expenses in a structured format
- Search expenses by category
- Monthly budget setting and warning system
- Category-wise expense analysis
- Graphical representation of expenses using charts
- Basic smart analysis and predictions
- Detection of recurring expenses

These features make the project more than just a basic expense tracker.

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6. System Architecture / Flow

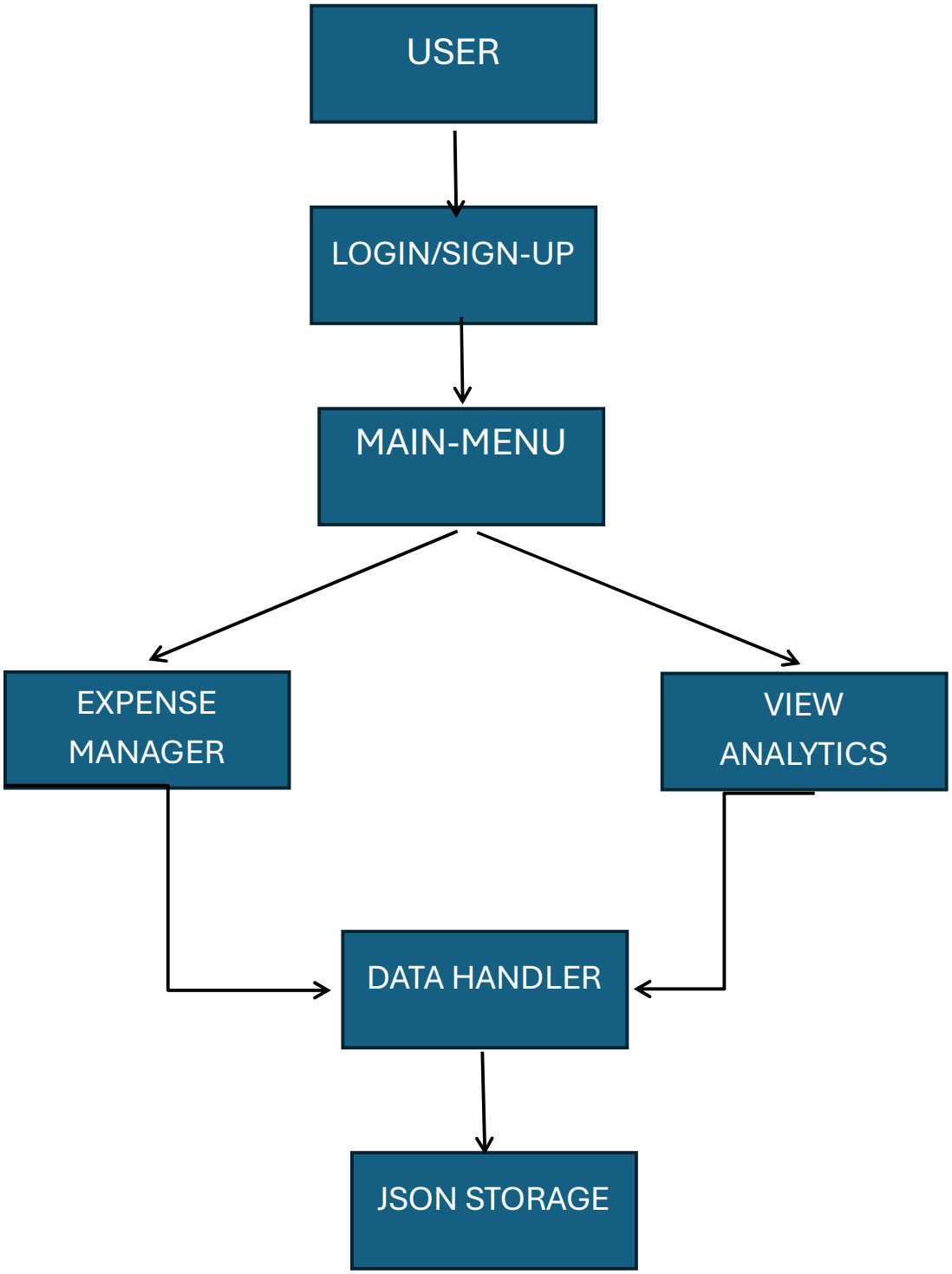
The overall flow of the system is simple:

User starts the program and logs in or signs up. After authentication, the user is presented with a menu containing different options.

Based on the selected option, the program performs operations like adding expenses, viewing data, or generating analysis.

All the data is stored in JSON files and updated in real time. The output is then displayed back to the user through the terminal.

Flow can be summarized as:



7. Technologies Used

The following technologies were used in this project:

- Python for core development
- JSON for data storage
- Matplotlib for graphs and visualization
- Hashlib for password security

The focus was more on logic building rather than using complex frameworks.

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8. Output Screenshots

```
====== 🚀 Welcome to SpendWise ======
1. Login
2. Signup
Enter choice: 1
Enter username: aviraj
Enter password: avi*123
❌ Invalid credentials

====== 🚀 Welcome to SpendWise ======
1. Login
2. Signup
Enter choice: 2
Enter username: avi
Enter password: avi@07
✅ Signup successful!
```

```
====== 💰 SpendWise (avi) ======
1. ➕ Add Expense
2. 📄 View Expenses
3. 🗑️ Delete Expense
4. ✏️ Update Expense
5. 💰 Total Spending
6. 📊 Category Total
7. 🔍 Search Expense
8. 🎯 Set Budget
9. 🚪 Logout
10. 📊 Category Chart
11. 📈 Spending Trend
12. 🧠 AI Spending Analysis
13. ❌ Exit App
Enter your choice:
```

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====== 🧠 AI SPENDING ANALYSIS ======

📊 Highest spending: Room Rent (₹3000.0)
📈 Room Rent takes 32.97% of your spending

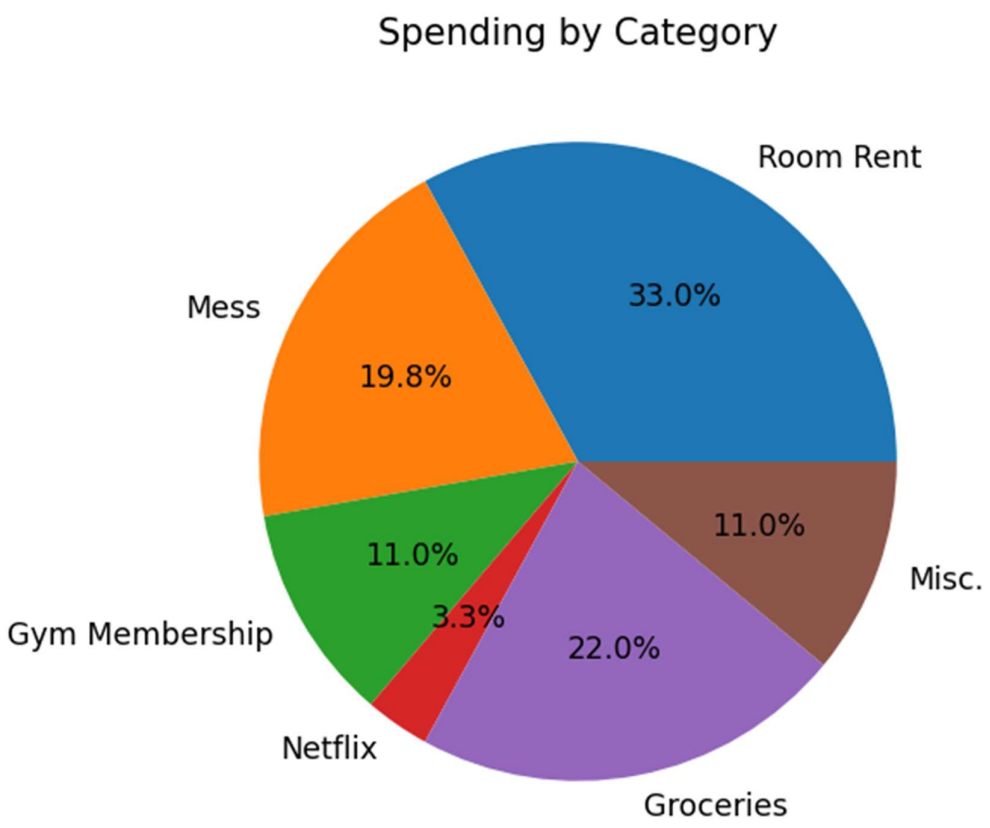
💡 Insights:
⚠️ Room Rent is your dominant expense
📉 Your spending is decreasing (good job!)

🔮 At this rate, you may spend around ₹272970 this month

🧠 Smart Suggestions:
💡 Monitor your Room Rent expenses for better savings

💯 Financial Health Score: 90/100

🔁 No recurring expenses detected
```



9. Challenges Faced

While building this project, some challenges I faced were:

- Managing separate data for different users
- Implementing a proper login system
- Handling file read/write operations without errors
- Designing logic for analysis and predictions
- Keeping the program simple while adding features

Some of these took time to figure out and required debugging multiple times.

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10. Limitations

The project also has some limitations:

- It is command-line based, so no graphical interface
- Data is stored locally and not online
- Analysis is basic and not very advanced
- Not suitable for large-scale usage

These can be improved in future versions.

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11. Future Scope

There are several improvements that can be made:

- Develop a graphical user interface (GUI)
- Convert it into a web application
- Use a proper database instead of JSON
- Implement real machine learning for better analysis
- Add features like data export and reports

This project can be expanded into a much more advanced system.

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12. Conclusion

SpendWise started as a simple idea but gradually developed into a more feature-rich application. It helped in understanding important programming concepts like file handling, modular coding, and user authentication.

The project also shows how basic logic can be used to create useful tools without relying on complex technologies.

Overall, it was a good learning experience and gave practical exposure to building real-world applications. Some parts can still be improved, but the current version works well for its purpose.